

DATA ANALYSES – ISBMM104 (UNamur)

TENTATIVE PROGRAM (Antoine CHRISTIAENS)

	MODULE	CONTENT	AT HOME PRE-REQUISIT	THEORY (IN-CLASS), <i>h</i>	HANDS-ON (IN-CLASS), <i>h</i>
1	INTRODUCTION	Type of data, variables, basis of R language and coding, importing database, datasets, reporting guidelines, open sciences principles	/	2	1
2	DATA MANAGEMENT (1)	Joins and merging, reshaping data, recoding variables, factor variables, dates/time variables, tidy data principles	/	1	2
3	DATA MANAGEMENT (2)	Data cleaning, identification of outliers, missing data exploration, handling missing values, common data manipulation workflows	/	1	2
4	DESCRIPTIVE STATISTICS AND EXPLORATORY ANALYSES	Exploratory data analysis, summary statistics, distributions and normality, contingency tables, exploratory subgroup analyses, reporting results in tables	Video (45 min): Descriptive statistics	1	2
5	DATA VISUALIZATION AND FIGURES	Type of graphics, Grammar of graphics, Exploratory vs explanatory figures	/	1	2
6	PERSONAL WORK (1)	In-class work on your personal analysis (1)	/	0	3
7	MULTIVARIABLE ANALYSES	Statistical inference, linear regression, logistic regression, model interpretation	Video (60 min): Statistical inference	1	2
8	PREDICTION (1)	Design and preparation of analysis, Model development, penalisation, performance, validation.	Video (30 min): prediction models	1	2
9	PREDICTION (2)	Machine learning, supervised and unsupervised learning, Dataset, decision tree, random forest	/	2	1
10	CLASSIFICATION	Clustering, finite mixture models	Video (30 min): Clustering and latent analyses	1	2
11	PERSONAL WORK (2)	In-class work on your personal analysis (2)	/	0	3
12	LONGITUDINAL ANALYSIS	Management of time and chronology, repeated measures, mixed models, trajectories, censoring, time-to-event analysis, survival analysis	/	1	2
13	CAUSAL APPROACHES	DAGs, propensity scores, IPTW	Video (30 min): Causal assumptions	2	1